

ZHUOYUN DU

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🎯 RESEARCH INTERESTS

1. **Autonomous Evolution in Multi-Agent Systems:** Transcending human-designed workflows to enable agents to forge **self-optimizing** coordination paradigms.
2. **Latent Space Communication & Reasoning:** Eliminating the autoregressive token bottleneck to establish efficient, high-dimensional **"machine-native"** interaction.
3. **Autonomous Evaluation:** Investigating how agents can establish **self-referential, emergent evaluation frameworks** in open-ended evolution without human-curated reward models.

🎓 EDUCATION

Zhejiang University, Hangzhou, China Sept 2024 – Jul 2027 (Expected)
M.Eng. in Data Science and Engineering Advisor: [Prof. Wei Chen](#) State Key Lab of CAD & CG

Jinan University, Guangzhou, China Sept 2020 – Jul 2024
B.Eng. in Computer Science and Technology (Rank: 3 / 99)

🧩 SELECTED PROJECT EXPERIENCE

Multi-Agent Collaboration via Cross-Team Orchestration November 2023 – June 2024
Leader

Proposed **Croto**, a scalable multi-team framework that addresses the single-outcome bottleneck in agent collaboration. By orchestrating parallel agent teams to explore diverse solutions and interact via cross-team mechanisms, Croto expands the solution space and outperforms ChatDev and MetaGPT. Paper accepted to Findings of [ACL 2025](#). [\[arXiv\]](#)

Agent Coevolution Framework for Standardized Patients August 2024 – December 2024
Leader

Proposed **EvoPatient**, a multi-agent coevolution framework for obtaining high-quality, usable LLM-simulated Standardized Patients without human supervision. The system autonomously generates, validates, and retains high-fidelity diagnostic trajectories through iterative self-improvement, achieving notable gains in requirement alignment and robustness. Paper accepted to [ACL 2025](#). [\[arXiv\]](#)

Enabling Agents to Communicate Entirely in Latent Space January 2025 – September 2025
Leader

Proposed a new paradigm for direct agent communication in **latent space**. By transmitting hidden states instead of natural language with efficient compression, it overcomes the token bottleneck and achieves up to **24× efficiency gain** with comparable performance. Paper accepted to [ACL 2026](#) (Oral recommended). [\[arXiv\]](#)

💼 WORK EXPERIENCE

ByteDance **Data Dept** **Talent Program Intern** March 2026 – Present

- **Core Research (LLM Post-training, Preference Alignment & Interactive Evaluation):** Designed and built an automated pipeline based on high-fidelity human-like agents. Simulated diverse user profiles and behavioral patterns to generate high-quality data supporting SFT and RLHF/DPO processes.

Alibaba **Future Living Lab** **Research Intern** January 2025 – March 2026

- **Core Research:** Focused on multi-agent reasoning and communication in latent space. Led a 9-month frontier research project around latent space communication, replacing traditional natural language with hidden states and leveraging efficient communication compression to boost multi-agent performance and efficiency in complex tasks.
- **Results:** Gained extensive experience in LLM fine-tuning and large-scale distributed multi-GPU (128+) training; authored two first-author papers during the internship ([Interlat in ACL](#) & [Online-PVLM](#)).

Tsinghua University **THUNLP** **Research Assistant** December 2023 – August 2024

- **Core Research:** Participated in multi-agent swarm intelligence research based on [ChatDev](#) in Prof. [Zhiyuan Liu](#)'s group, exploring the enhancement of emergent collective capabilities through cooperation and competition mechanisms among LLMs.
- **Results:** Core contributor and project lead for two major branches: **Croto (ACL 2025)** and **MACNet (ICLR 2025)**. Primarily led the development of Croto, and contributed to agent information integration, profile generation, and

paper drafting for MACNet. Related open-source projects have accumulated over **32k stars** on GitHub.

⚙️ RELATED SKILLS

- **LLM & Machine Learning:** Multi-Agent System architecture design, multi-GPU training experience (128+ GPUs); SFT, LoRA; familiar with training acceleration technologies like DeepSpeed and Flash Attention.
- **Programming Languages & Frameworks:** Python, C/C++, Bash, Git, $\LaTeX$; PyTorch, Hugging Face ecosystem (Transformers, Datasets).
- **Languages:** English (IELTS: 7.5); Mandarin (Native).

🏆 AWARDS AND HONORS

- 2025 **National Scholarship**, Outstanding Graduate Student Scholarship, **3 Undergraduate Scholarships**, **First-Class Scholarship for Outstanding Graduates**.

🔥 INTERESTS AND SPECIALTIES

Broad academic vision and strong curiosity, with great collaboration and leadership skills in complex cross-team co-operation. Diverse hobbies, former **National Champion in Amateur Épée Fencing**, hold a Grade 7 Piano Certificate, and passionate about snowboarding, photography, and various arts and cultural activities.

📖 PUBLICATIONS

[†] indicates co-first / equal contribution.

First-author / Co-first-author

1. **Zhuoyun Du**[†], Runze Wang[†], Huiyu Bai, Zouying Cao, Xiaoyong Zhu, Bo Zheng, Wei Chen and Haochao Ying. Enabling Agents to Communicate Entirely in Latent Space. *ACL 2026*. [\[paper\]](#) [\[Code\]](#)
2. **Zhuoyun Du**[†], Lujie Zheng[†], Renjun Hu, Yuyang Xu, Xiawei Li, Ying Sun, Wei Chen, Jian Wu, Haolei Cai, and Haochao Ying. LLMs can simulate standardized patients via agent coevolution. *ACL 2025*. [\[paper\]](#) [\[Code\]](#)
3. **Zhuoyun Du**[†], Chen Qian[†], Wei Liu, Zihao Xie, Yifei Wang, Yufan Dang, Weize Chen, and Cheng Yang. Multi-agent software development through cross-team collaboration. *Findings of ACL 2025*. [\[paper\]](#) [\[Code\]](#)
4. Huiyu Bai[†], Runze Wang[†], **Zhuoyun Du**[†], Yiyang Zhao, Fengji Zhang, Haoyu Chen, Xiaoyong Zhu, Bo Zheng and Xuejiao Zhao. Online-PVLM: Advancing Personalized VLMs with Online Concept Learning. *arXiv preprint arXiv:2511.20056*, 2025. [\[arXiv\]](#)

Co-authored

5. Shan He, Runze Wang, **Zhuoyun Du**, Huiyu Bai, Zouying Cao, Yucheng, Bo Zheng. Learning to Evolve: A Self-Improving Framework for Multi-Agent Systems via Textual Parameter Graph Optimization. *Findings of ACL 2026*. [\[paper\]](#)
6. Yuewen Liu, Peng Xu, **Zhuoyun Du**, Muxi Diao, Anyi Zhang, Yang Li, Yutong Zhang. MemCoRL: Alternating Co-Optimization of Memory Retrieval and Utilization via Collaborative Reinforcement Learning. *ACL 2026*
7. Chen Qian[†], Zihao Xie[†], Yifei Wang[†], Wei Liu, Yufan Dang, **Zhuoyun Du**, Weize Chen, Cheng Yang, Zhiyuan Liu, and Maosong Sun. Scaling large-language-model-based multi-agent collaboration. *ICLR 2025*. [\[paper\]](#)
8. Wei Liu[†], Chenxi Wang[†], Yifei Wang, Zihao Xie, Rennai Qiu, Yufan Dang, **Zhuoyun Du**, Weize Chen, Cheng Yang, and Chen Qian. Autonomous agents for collaborative task under information asymmetry. *NeurIPS 2024*. [\[paper\]](#)
9. Yuyang Xu, Yi Cheng, Haochao Ying, **Zhuoyun Du**, Renjun Hu, Xing Shi, Wei Lin and Jian Wu. SSPO: Self-traced Step-wise Preference Optimization for Process Supervision and Reasoning Compression. [\[arXiv\]](#)